

ECON 311

Lecture 12: Introduction to Short Run

Fisher Equation

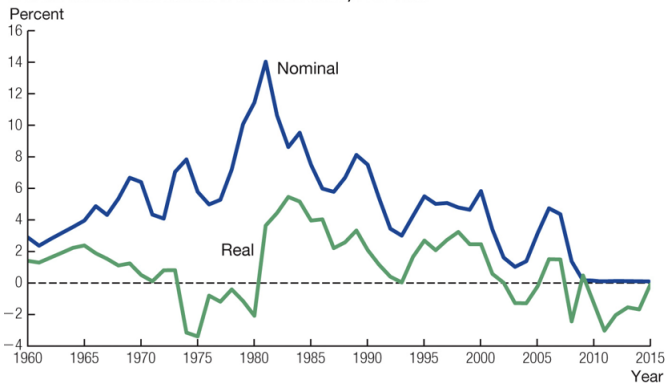
- The Fisher Equation allows us to convert nominal returns on investment into real returns:

$$i = R + \Pi \Rightarrow R = i - \Pi$$

- ▶ i : nominal interest rate
 - ▶ R : real interest rate
 - ▶ Π : inflation rate
- R depends on the MPK , which is usually stable in the short-run
 - ▶ Therefore, the nominal interest rate is usually high when inflation is high

Fisher Equation

Real and Nominal Interest Rates in the United States, 1960–2015



Short-Run Macroeconomics

The Goods Market in the Short Run

Appendix

Short-Run Macroeconomics

- The long-run model is a guide to how the economy behaves on average
 - ▶ Determines potential output and long-run inflation
- Potential output
 - ▶ Amount of production if all inputs were utilized at their long-run sustainable levels
- At any given time, the economy is unlikely to exactly equal the long-run average
- The short-run model:
 - ▶ Determines current output and current inflation

Short-Run Macroeconomics

- Actual output is equal to the long-run trend plus short-run fluctuations:

$$\underbrace{\text{actual output}}_{Y_t} = \underbrace{\text{long-run trend}}_{\bar{Y}_t} + \underbrace{\text{short-run fluctuations}}_{\text{function of } \tilde{Y}_t}$$

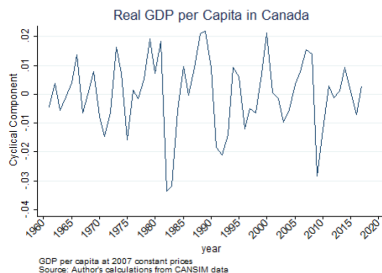
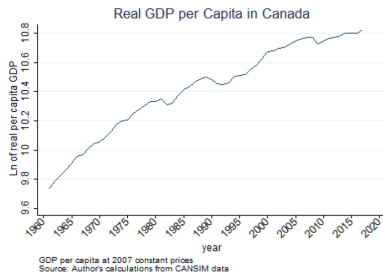
- ▶ Long-run trend: potential output
- ▶ Short-run fluctuations: deviations from $\bar{Y}_t$

Short-Run Macroeconomics

- “De-trended output” or short-run output:
 - ▶ The difference in actual and potential output, expressed as a percentage of potential output

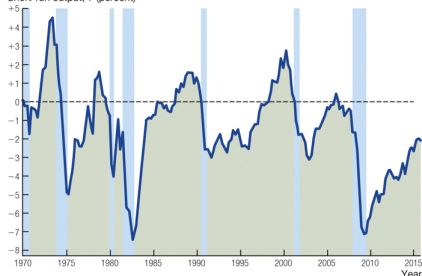
$$\tilde{Y}_t = \frac{Y_t - \bar{Y}_t}{\bar{Y}_t}$$

Short-Run Macroeconomics

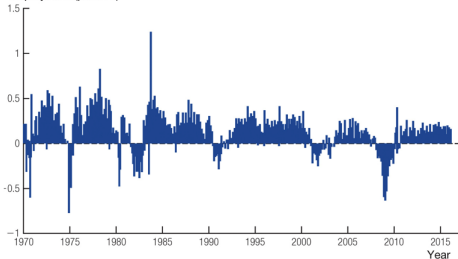


Economics Fluctuations and Unemployment

U.S. Economic Fluctuations, 1970–2015
Short-run output, $\tilde{Y}$ (percent)



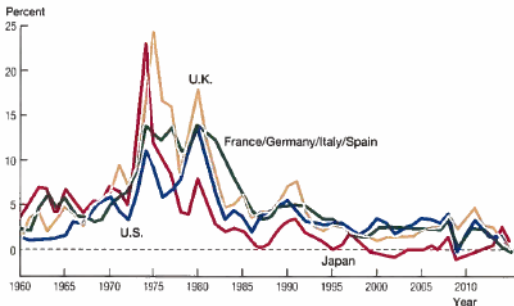
Changes in U.S. Employment, 1970–2015
Monthly change in employment (percent)



- Recessions are very costly:
 - ▶ E.g. during a typical recessions GDP per households is around \$12,000 below potential output
 - ▶ Unemployment has long-run negative effects on the population.

Inflation in the Short Run

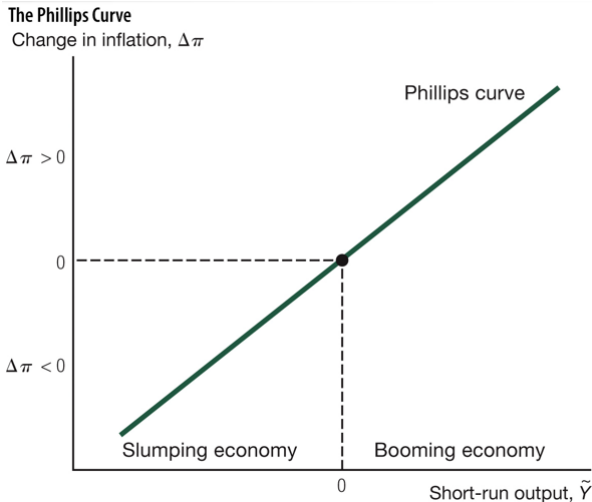
Inflation Rates in Certain Rich Countries



- Inflation increases during economic booms
- Inflation decreases during recessions

Output-Inflation Trade-off

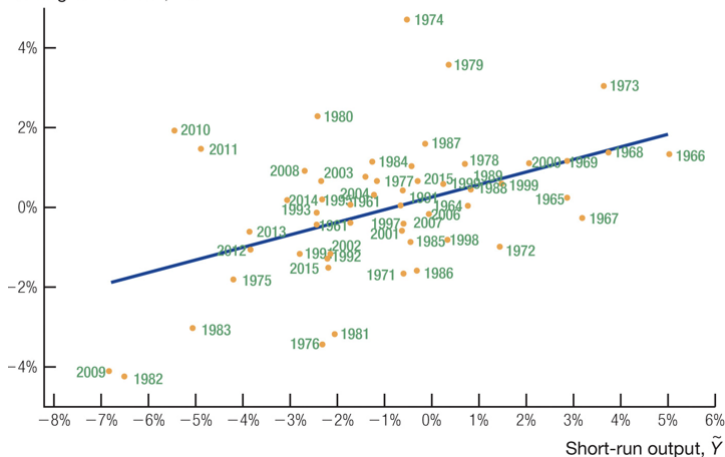
- The Phillips Curve denotes the output-inflation trade-off:



Phillips Curve in the US

Measuring the Phillips Curve, 1960–2015

Change in inflation, $\Delta\pi$



The Basis of the Short-Run Model

- The economy is constantly being hit by shocks that cause fluctuations in economic activity
 - ▶ Positive shocks increase economic activity beyond its natural rate, which results in inflation.
 - ▶ Negative shocks decrease economic activity below its natural rate, which results in recessions.
- Government through fiscal and monetary policy attempts to neutralize the effects of the shocks.
- This creates a constant cycle of boom and bust.
 - ▶ The duration and severity depends on the shock and government's reaction to the shock.

Economic Activity and Unemployment

- The Short-Run Model analyzes fluctuations in output and not in unemployment
- There is a direct connection between unemployment and output, known as Okun's Law:

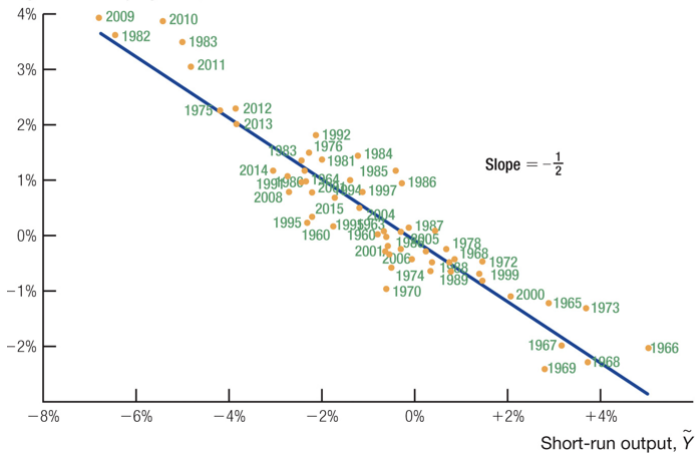
$$u - \bar{u} = -\frac{1}{2} \tilde{Y}$$

- ▶ u : current rate of unemployment
- ▶ $\bar{u}$: natural rate of unemployment
- ▶ $\tilde{Y}$: short-run output

Okun's Law in the US

Okun's Law for the U.S. Economy, 1960–2015

Cyclical unemployment, $u - \bar{u}$



Preview of the Short-Run Model

In the remainder of the semester:

1. We will carefully derive the IS-MP model and study the effects of shocks and government stabilization policy in this framework.
2. Afterward, we will study the connection between the IS-MP model and the Phillips curve.
3. Finally, we will put all the parts together and derive a model of the short-run economy, AS-AD, that simulates the periods of boom and bust that characterize modern economies.

Short-Run Macroeconomics

The Goods Market in the Short Run

Appendix

Introduction

- Fluctuations in output around its potential level arise from aggregate demand variations.
- In this lecture, I introduce a simple model of aggregate demand to study how output is determined:
 - ▶ There are two key variables:
 - ▶ Aggregate expenditure= $C + I + G + X - M$
 - ▶ Aggregate income= Y
 - ▶ In equilibrium, expenditure equals income:

$$Y = C + I + G + X - M$$

Components of aggregate expenditure

1. **Consumption:** Modelled as a function of income:

$$C = a_c + b_c Y \quad , \quad b_c < 1 \text{ marginal propensity to consume}$$

2. **Investment:** Modelled as a function of nominal interest rates:

$$I = a_i - b_i(R) \quad , \quad b_i < 1$$

3. **Government Expenditure:** Modelled as exogenous: $G = \bar{G}$

4. **Net Exports:** Modelled as exogenous: $NX = \bar{NX} = \bar{X} - \bar{M}$

Equilibrium in the Goods Market

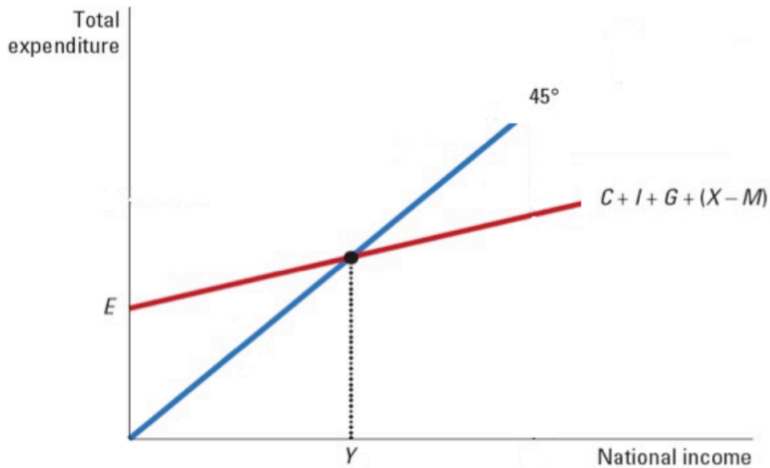
- In equilibrium, Agg expenditure=Agg. output

$$Y = C + I + G + X - M$$

$$\Rightarrow Y = a_c + b_c Y + a_i - b_i(R) + \bar{G} + \bar{X} - \bar{M}$$

$$\Rightarrow Y = \underbrace{\left[\frac{1}{1 - b_c} \right]}_{\text{multiplier}} * [a_c + a_i - b_i(R) + \bar{G} + \bar{X} - \bar{M}]$$

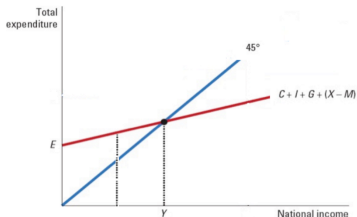
The Keynesian Cross



Equilibrium in the Goods Market – Revisited

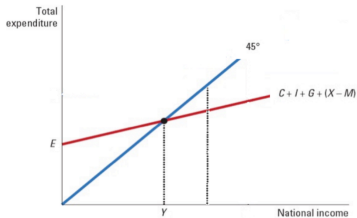
When $E > Y$, there is excess demand

- Firms increase production
- Excess demand falls



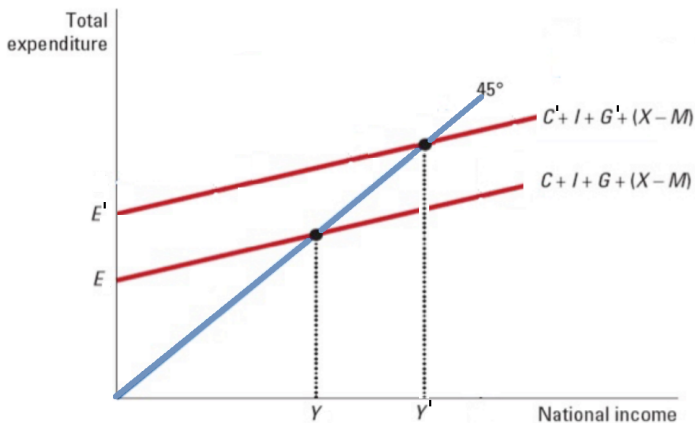
When $E < Y$, there is excess supply

- Firms decrease production
- Excess supply falls



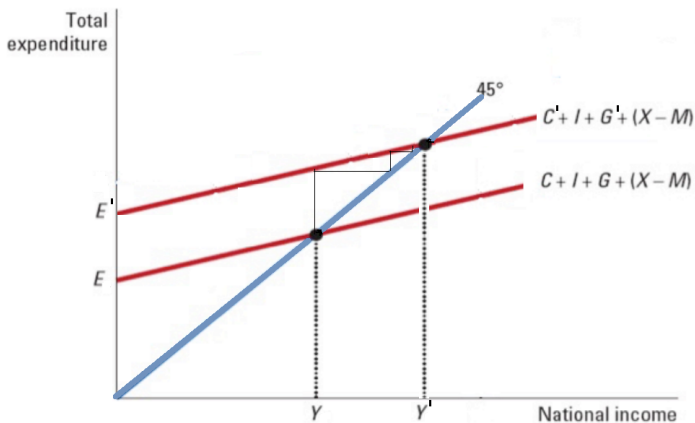
Changes in expenditure and the multiplier

Exogenous increase in G from $\bar{G}$ to $\bar{G}'$



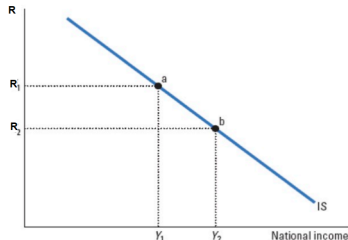
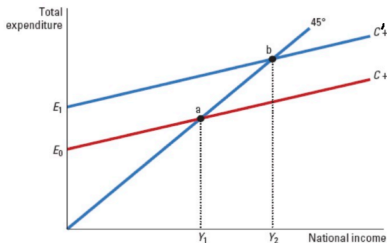
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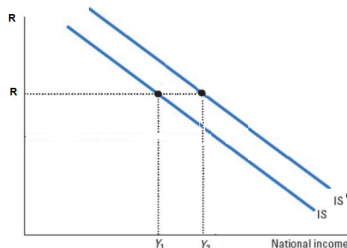
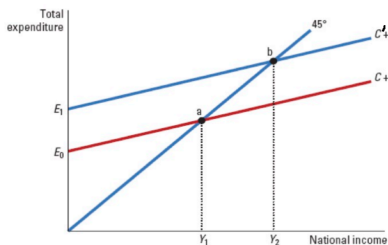
Deriving the IS-curve

$$\text{IS curve: } Y = \left[\frac{1}{1-b_c} \right] * \left[a_c + a_i - b_i(R) + \bar{G} + \bar{X} - \bar{M} \right]$$



Parametric Shifts in the IS-curve

$$\text{IS curve: } Y = \left[\frac{1}{1-b_c} \right] * \left[a_c + a_i - b_i(R) + \bar{G} + \bar{X} - \bar{M} \right]$$



Change of scale

- We are interested in $\tilde{Y}_t = \frac{Y_t - \bar{Y}}{\bar{Y}}$, so we are going to rewrite the equilibrium as follows:

$$\tilde{Y}_t = \left[\frac{1}{1 - b_c} \right] * [\bar{a} - \bar{b}_i(R_t - \bar{r})] - 1$$

$$\blacktriangleright \bar{a} = \frac{a_c}{\bar{Y}} + \frac{a_i}{\bar{Y}} - \frac{b_i}{\bar{Y}}\bar{r} + \frac{\bar{G}}{\bar{Y}} + \frac{\bar{X}}{\bar{Y}} - \frac{\bar{M}}{\bar{Y}}$$

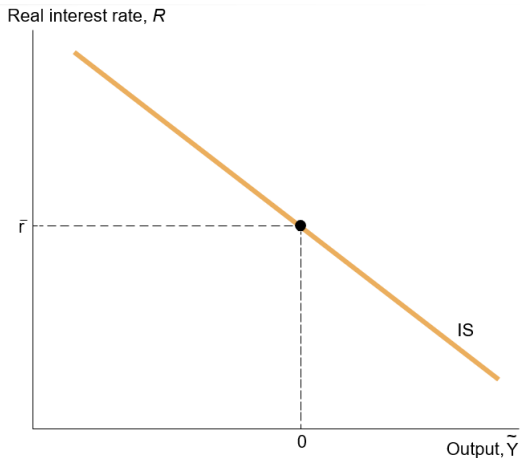
$$\blacktriangleright \bar{b}_i = \frac{b_i}{\bar{Y}}$$

- If all exogenous variables are at their long-run equilibrium:

$$\left[\frac{1}{1 - b_c} \right] * \bar{a} = 1 \Rightarrow \tilde{Y}_t = \left[\frac{1}{1 - b_c} \right] [-\bar{b}_i * (R_t - \bar{r})]$$

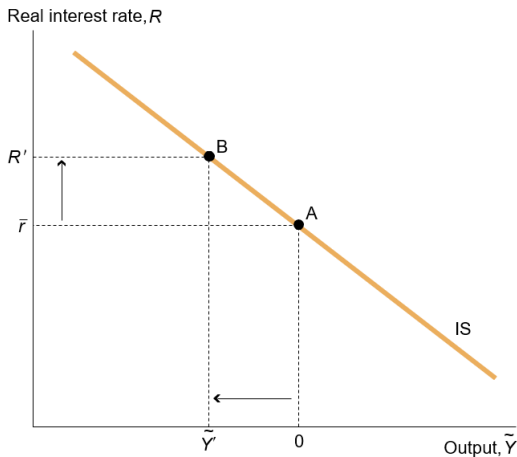
IS-curve with new scale

$$\text{IS-curve: } \tilde{Y}_t = \left[\frac{1}{1-b_c} \right] * \left[\bar{a} - \bar{b}_i(R_t - \bar{r}) \right] - 1, \quad \bar{a} \left[\frac{1}{1-b_c} \right] = 1$$



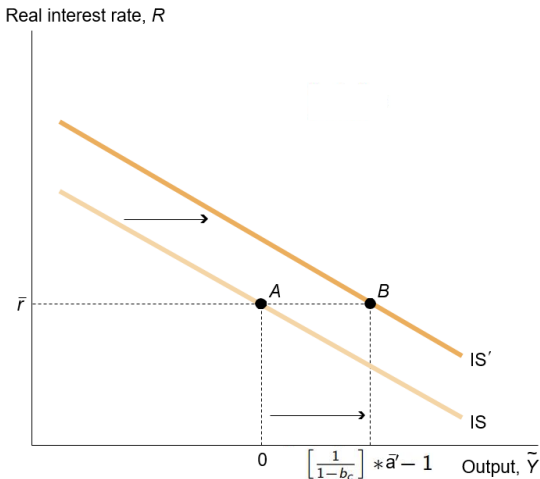
Changes in the real interest rate

$$\text{IS-curve: } \tilde{Y}_t = \left[\frac{1}{1-b_c} \right] * \left[\bar{a} - \bar{b}_i(R_t - \bar{r}) \right] - 1, \quad \bar{a} \left[\frac{1}{1-b_c} \right] = 1$$



Temporary Aggregate Demand Shock

$$\text{IS-curve: } \tilde{Y}_t = \left[\frac{1}{1-b_c} \right] * [\bar{a} - \bar{b}_i(R_t - \bar{r})] - 1, \quad \bar{a} \left[\frac{1}{1-b_c} \right] = 1, \quad \bar{a}' > (1 - b_c)$$

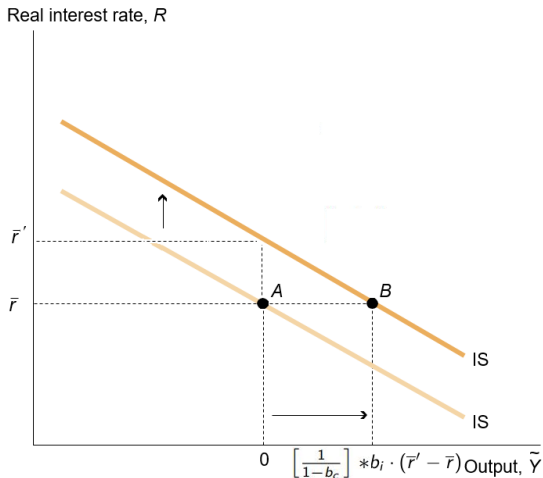


Permanent Aggregate Shock

- Permanent shocks impact potential output and originate from the production side of the economy, which have two effects:
 1. Permanent shocks affect demand parameters, which affect both long- and short-run output levels equally.
 2. Permanent production shocks affect the MPK
 - ▶ This has no effect in the long-run because $\bar{r} = MPK$
 - ▶ This results in output overshooting in the SR as the economy adjusts to its new potential output.

E.g. Permanent TFP shock

$$\text{IS-curve: } \tilde{Y}_t = \left[\frac{1}{1-b_c} \right] * [\bar{a} - \bar{b}_i(R_t - \bar{r})] - 1, \quad \bar{a} \left[\frac{1}{1-b_c} \right] = 1, \quad \bar{r}' > \bar{r}$$



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Change of Scale: Derivation (part 1)

- We begin with the solution to Y_t in the SR equilibrium:

$$Y_t = \frac{1}{1 - b_c} (a_c + a_i - b_i R_t + \bar{G} + \bar{X} - \bar{M})$$

- We first divide both sides by the long-run output level $\bar{Y}$ and then subtract one from both sides:

$$\underbrace{\frac{Y_t}{\bar{Y}} - 1}_{\tilde{Y}_t} = \frac{1}{1 - b_c} \left(\frac{a_c}{\bar{Y}} + \frac{a_i}{\bar{Y}} - \frac{b_i}{\bar{Y}} R_t + \frac{\bar{G}}{\bar{Y}} + \frac{\bar{X}}{\bar{Y}} - \frac{\bar{M}}{\bar{Y}} \right) - 1$$

- We add and subtract $\frac{b_i}{\bar{Y}} \bar{r}$:

$$\tilde{Y}_t = \frac{1}{1 - b_c} \left(\frac{a_c}{\bar{Y}} + \frac{a_i}{\bar{Y}} - \frac{b_i}{\bar{Y}} R_t + \frac{b_i}{\bar{Y}} \bar{r} - \frac{b_i}{\bar{Y}} \bar{r} + \frac{\bar{G}}{\bar{Y}} + \frac{\bar{X}}{\bar{Y}} - \frac{\bar{M}}{\bar{Y}} \right) - 1$$

Change of Scale: Derivation (part 2)

- We rearrange terms to obtain:

$$\tilde{Y}_t = \frac{1}{1-b_c} [\bar{a} - \bar{b}_i (R_t - \bar{r})] - 1$$

where,

$$\bar{a} = \frac{a_c}{\bar{Y}} + \frac{a_i}{\bar{Y}} - \frac{b_i}{\bar{Y}} \bar{r} + \frac{\bar{G}}{\bar{Y}} + \frac{\bar{X}}{\bar{Y}} - \frac{\bar{M}}{\bar{Y}}$$

and,

$$\bar{b}_i = \frac{b_i}{\bar{Y}}$$

$$\text{and, } \bar{Y} = \frac{1}{1-b_c} \left(a_c^{LR} + a_i^{LR} - b_i^{LR} \bar{r} + \bar{G}^{LR} + \bar{X}^{LR} - \bar{M}^{LR} \right)$$

- From the previous equation, we can see that if all parameters in the SR are equal their long-run levels:

$$\bar{a} = \frac{a_c^{LR} + a_i^{LR} - b_i^{LR} \bar{r} + \bar{G}^{LR} + \bar{X}^{LR} - \bar{M}^{LR}}{\frac{1}{1-b_c} \left(a_c^{LR} + a_i^{LR} - b_i^{LR} \bar{r} + \bar{G}^{LR} + \bar{X}^{LR} - \bar{M}^{LR} \right)}$$

- Solving for this results in: $\bar{a} = 1 - b_c$